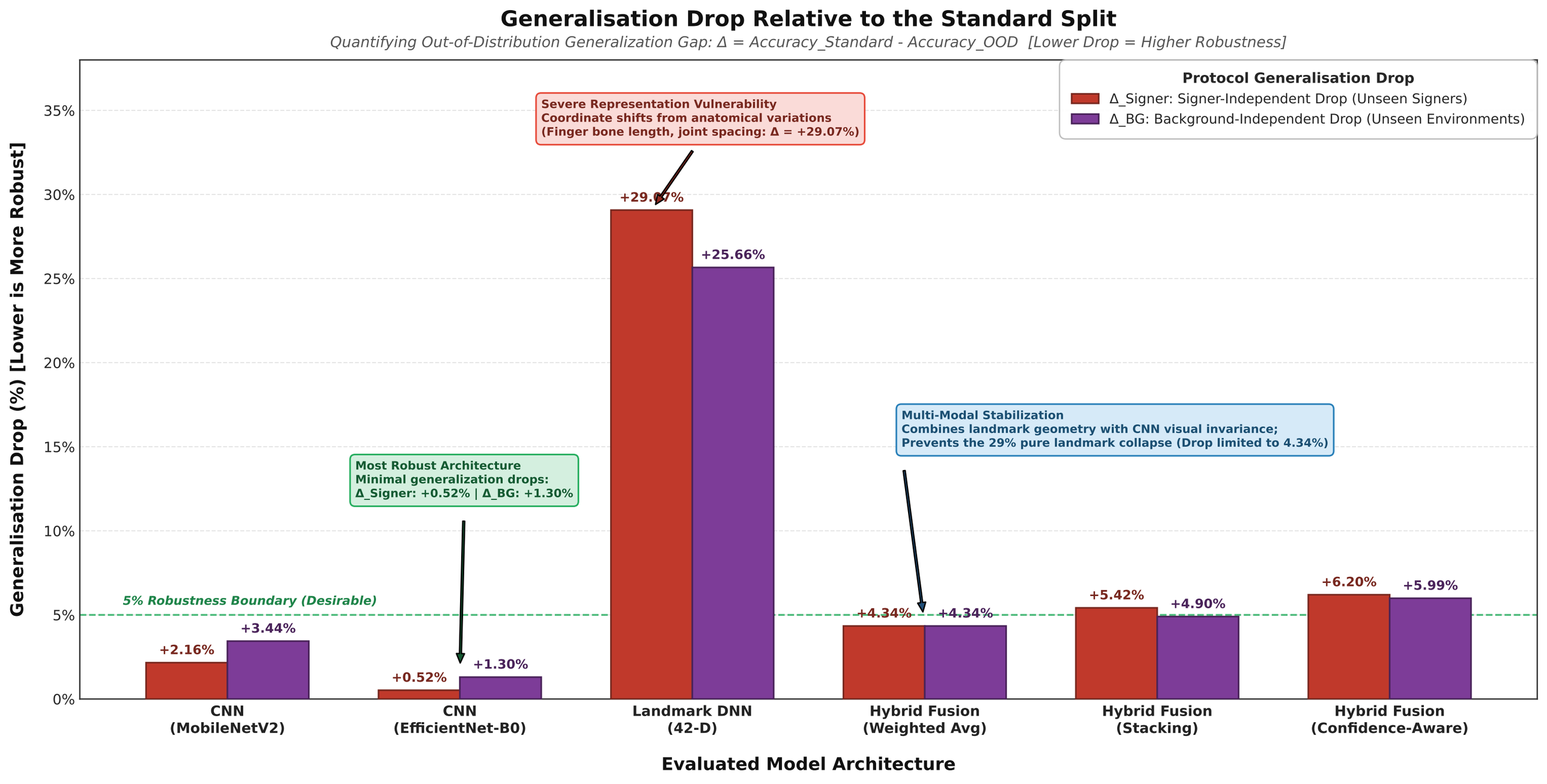

Generalisation Drop Relative to the Standard Split
Quantifying Out-of-Distribution Generalization Gap: Δ = Accuracy_Standard - Accuracy_OOD [Lower Drop = Higher Robustness]
Protocol Generalisation Drop
Δ_Signer: Signer-Independent Drop (Unseen Signers)
Δ_BG: Background-Independent Drop (Unseen Environments)
Severe Representation Vulnerability
Coordinate shifts from anatomical variations
(Finger bone length, joint spacing: Δ = +29.07%)
Most Robust Architecture
Minimal generalization drops:
Δ_Signer: +0.52% | Δ_BG: +1.30%
Multi-Modal Stabilization
Combines landmark geometry with CNN visual invariance;
Prevents the 29% pure landmark collapse (Drop limited to 4.34%)
5% Robustness Boundary (Desirable)
+2.16%
+3.44%
+0.52%
+1.30%
+29.07%
+25.66%
+4.34%
+4.34%
+5.42%
+4.90%
+6.20%
+5.99%
0%
5%
10%
15%
20%
25%
30%
35%
Generalisation Drop (%) [Lower is More Robust]
CNN (MobileNetV2)
CNN (EfficientNet-B0)
Landmark DNN (42-D)
Hybrid Fusion (Weighted Avg)
Hybrid Fusion (Stacking)
Hybrid Fusion (Confidence-Aware)
Evaluated Model Architecture